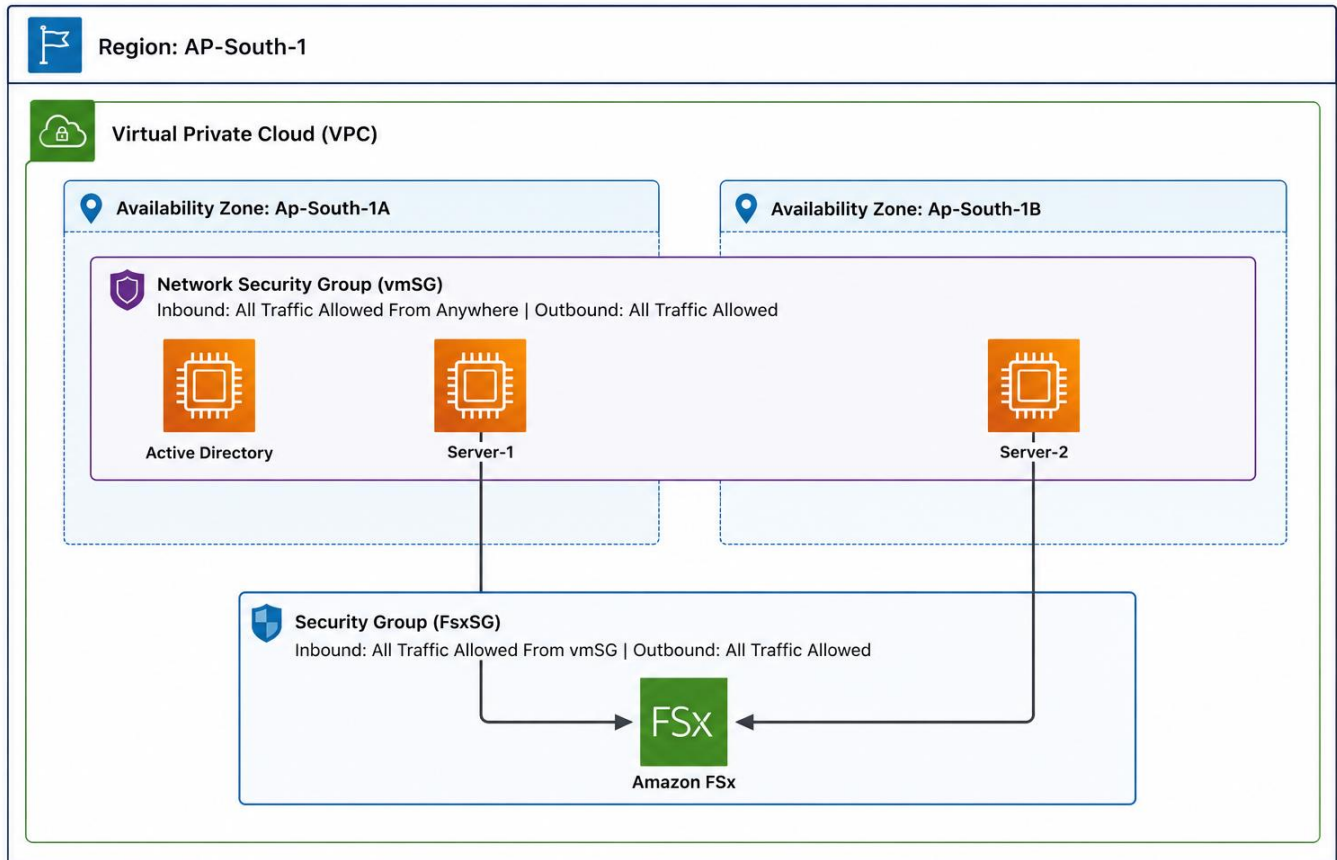


Amazon FSx: -

Summary: -

In this lab, I created multiple Windows EC2 instances and configured one server as an Active Directory server. Then I connected the other servers to the domain and created Amazon FSx shared storage. Finally, I mounted the shared drive on different servers so files could be accessed and shared between them.

Let's Create: -



- Now we need Two Security group (vmSG and FsxSG).
- Firstly we create vmSG
- Open aws console and search EC2 and in EC2 go to Security group and create.

Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info
All traffic ▼	All	All	A... ▼	0.0.0.0/0
				0.0.0.0/0 ✕
Add rule				

- 1st security group vmSG inbound rules.

Security Groups (1) [Info](#) [Actions](#) [Export security groups to CSV](#) [Create security group](#)

Find security groups by attribute or tag

vmSG [Clear filters](#)

Name	Security group ID	Security group name	VPC ID
-	sg-006f3303bb0b7b078	vmSG	vpc-072c4acd761c3b942

- Security group is created.
- Now we create 3 EC2 instance.
- 2 in ap-south-1a and 1 in ap-south-1b

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-072c4acd761c3b942 (default) [Refresh](#)

172.31.0.0/16

Subnet [Info](#)

subnet-04ffa8e5e9c79ad3d [Refresh](#)

VPC: vpc-072c4acd761c3b942 Owner: 463646775279
Availability Zone: ap-south-1a (aps1-az1) Zone type: Availability Zone
IP addresses available: 4090 CIDR: 172.31.32.0/20

[Create new subnet](#)

Auto-assign public IP [Info](#)

Enable

[Additional charges apply](#) when outside of [free tier allowance](#)

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)

Select security groups

[Compare security group rules](#)

vmSG [sg-006f3303bb0b7b078](#) [X](#)
VPC: vpc-072c4acd761c3b942

- 1st EC2 instance window 2016, t3.micro.
- And it's the network setting Subnet – ap-south-1a, security group – vmSG
- We need to create 2 Ec2 in ap-south-1a so create second with same things.
- Then create third Ec2 in ap-south-1b and Security group – vmSG.

Instances (4) [Info](#) Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets Choose filter set Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4
Server2	i-075e85303a3aceb37	Running	t3.micro	Initializing	ap-south-1b	ec2-13-12f
InstnaceToOne	i-08e22621356756fb8	Terminated	t2.medium	-	ap-south-1a	-
Server1	i-025b29cc62fd48a8b	Running	t3.micro	3/3 checks passed	ap-south-1a	ec2-13-20f
AD-Server	i-0152e27f6aa9f4045	Running	t3.micro	3/3 checks passed	ap-south-1a	ec2-43-20f

- All three instance is created.
- Right now the first thing is we need to configure ad server okay.

- So we are going to access ad server.
- First I'm going to access this using RDP.
- Now I want to change IP address
- You know, the server IP can be changed if I am going to restart.
- And this will be problematic.
- But whatever IP address I am getting as a default IP, I want to provide same IP as a static.
- Open cmd in your virtual Machine.

Administrator: C:\Windows\system32\cmd.exe

```
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

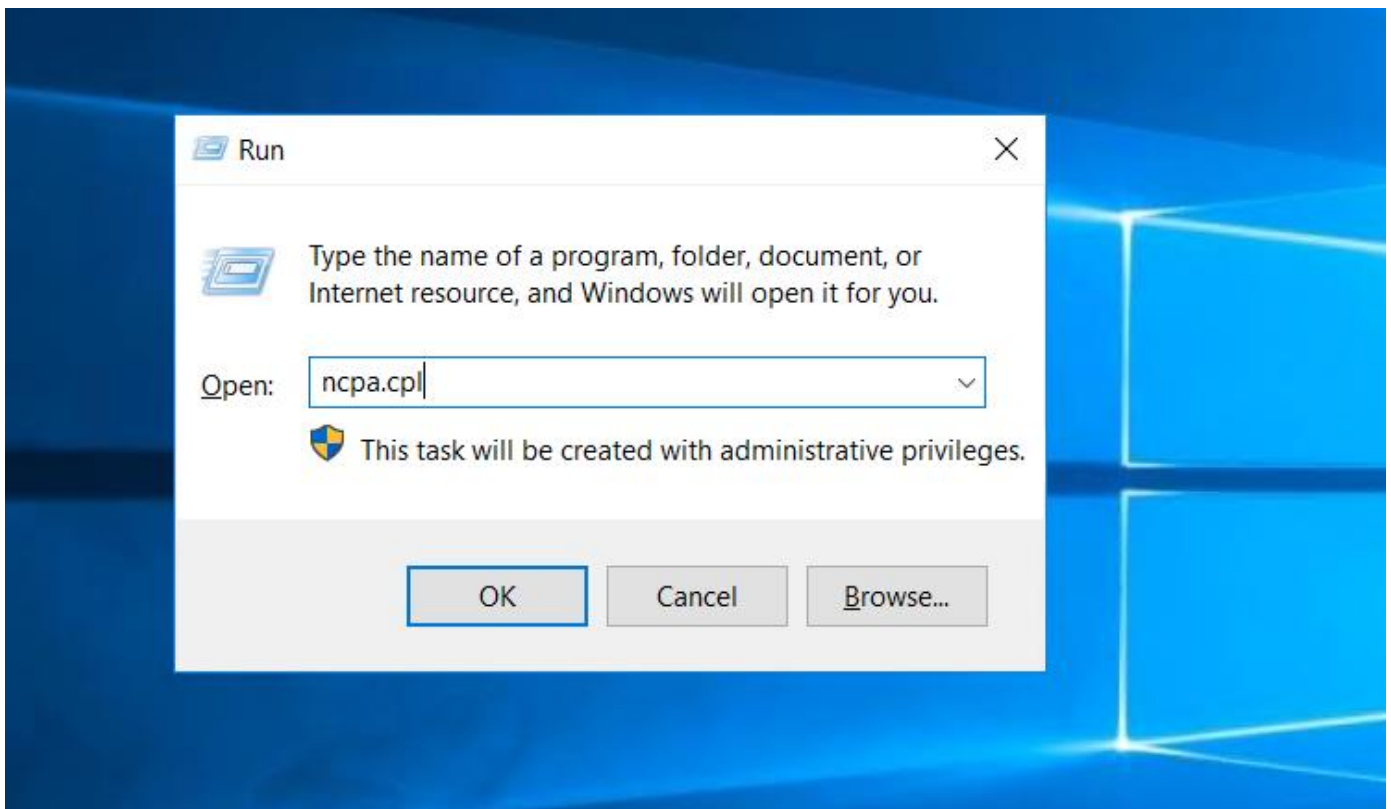
C:\Users\Administrator>ipconfig/all_
```

- Enter ipconfig/all

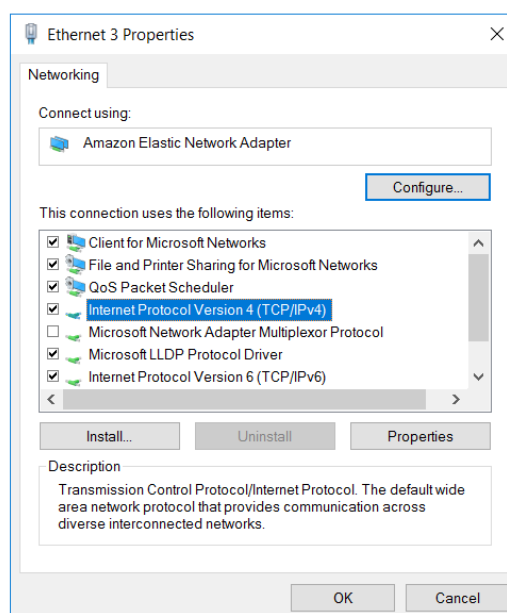
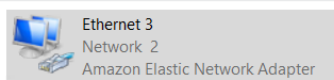
Ethernet adapter Ethernet 3:

```
Connection-specific DNS Suffix  . : ap-south-1.compute.internal
Description . . . . . : Amazon Elastic Network Adapter
Physical Address. . . . . : 02-A4-96-89-93-4F
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::654f:ecf3:c741:6ba8%4(Preferred)
IPv4 Address. . . . . : 172.31.42.0(Preferred)
Subnet Mask . . . . . : 255.255.240.0
Lease Obtained. . . . . : Thursday, May 21, 2026 8:40:50 AM
Lease Expires . . . . . : Thursday, May 21, 2026 10:10:51 AM
Default Gateway . . . . . : 172.31.32.1
DHCP Server . . . . . : 172.31.32.1
DHCPv6 IAID . . . . . : 134938601
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-A0-5F-EB-12-57-DE-AF-09-E9
DNS Servers . . . . . : 172.31.0.2
NetBIOS over Tcpip. . . . . : Enabled
```

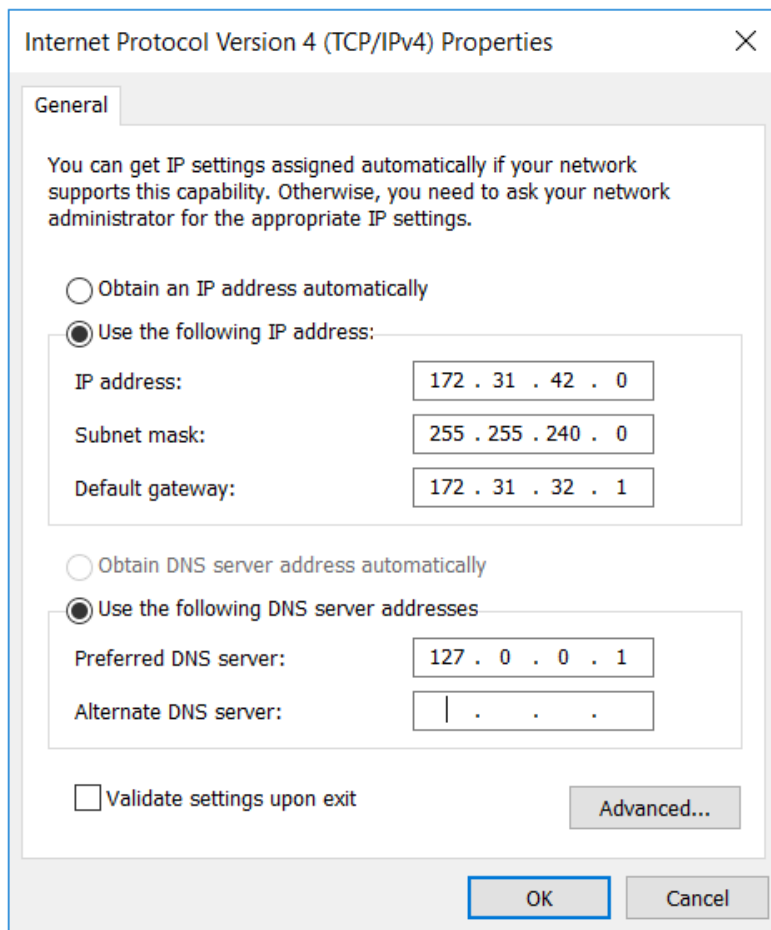
- Now we need these things.
- So minimize CMD.
- And press windows + R.



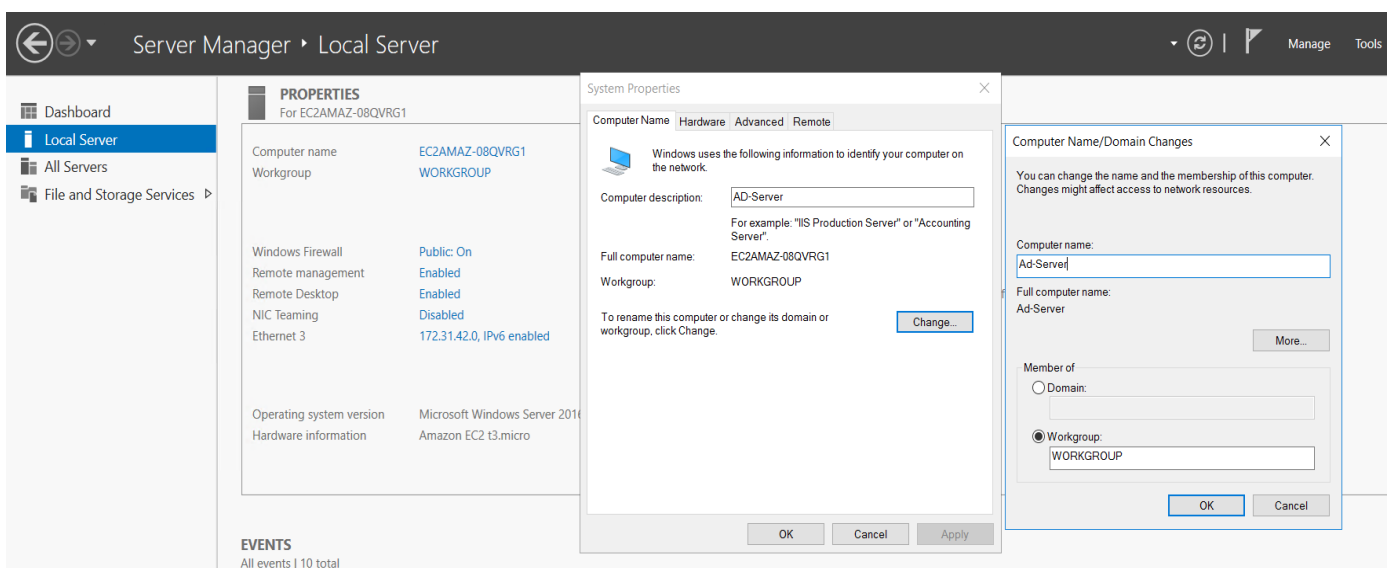
- Now Enter ncpa.cpl and press ok.



- Now right click on Ethernet 3 and select properties.
- Then select Internet protocol Version 4 (TCP/IPv4)

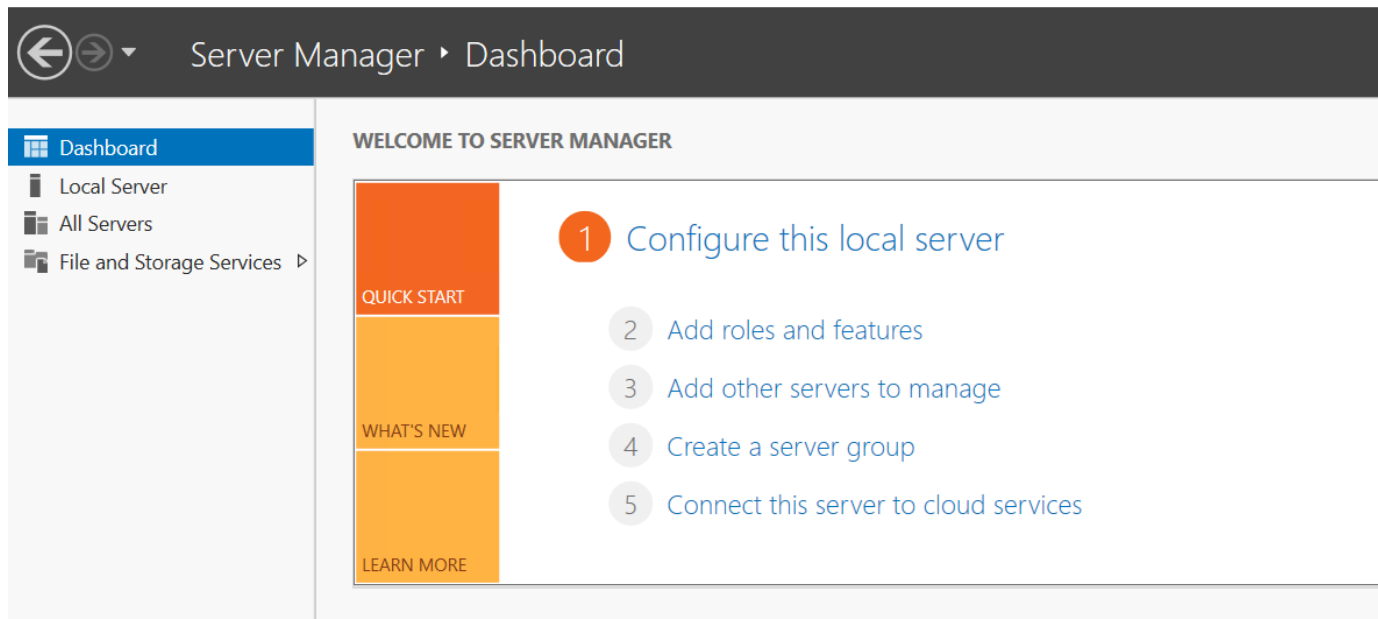


- Now one by one enter data from CMD.
- And in preferred DNS server 127.0.0.1
- Now when you will click okay it will disconnect for two seconds and then it will connect.
- If you have entered the right IP.
- If you have entered the wrong IP then you will lost your EC2 instance.
- In this case the process will be different.
- So just delete this and recreate this.
- Now open Server Manager.

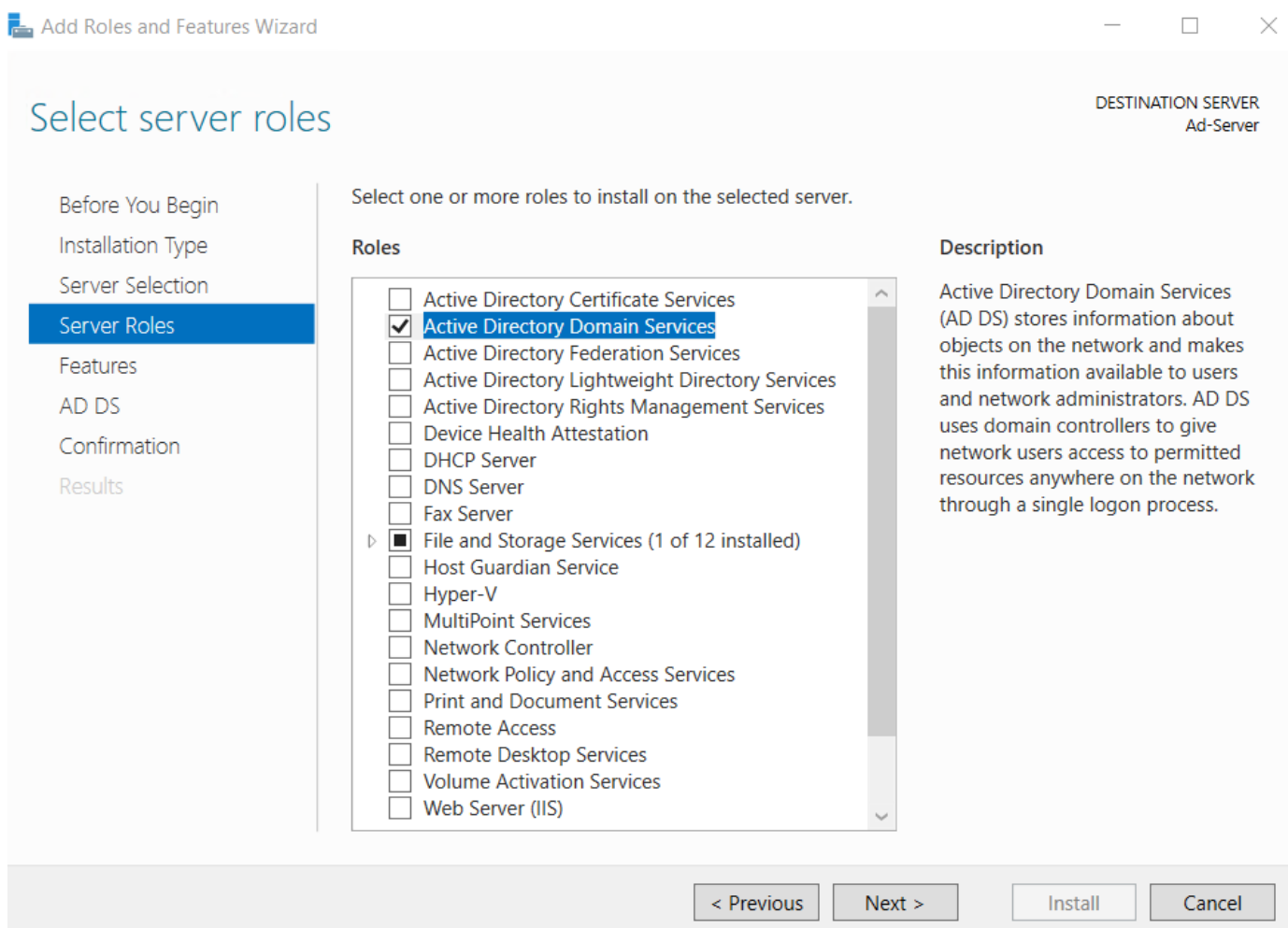


- Now go to Local server and change computer name as AD-Server.
- And click ok then it gives restart option restart it.
- Now open other two Ec2 instance using RDP.

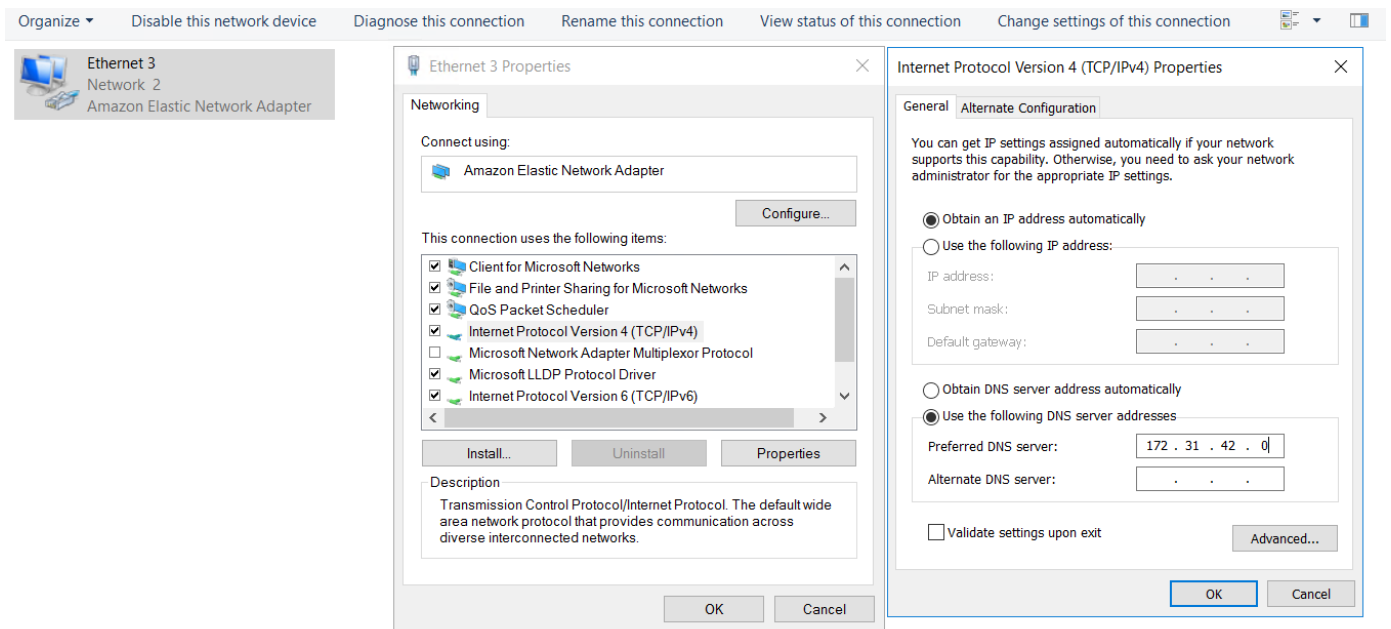
- Now go to AD-Server machine and open Server Manager.



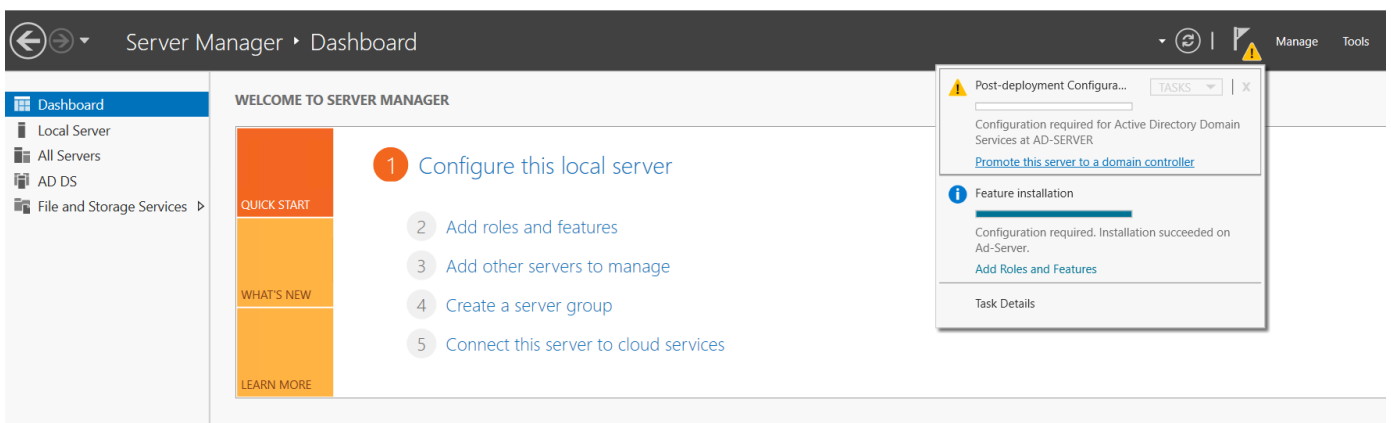
- Now click Add roles and features.



- Now click Next and in Server roles add Active Directory Domain Services.
- Then click Next and at last Install.
- Now we need to add Ad-Server Ip address that we get in CMD we need to add in Server1 and Server2 machines.



- Press window + R And Enter `ncpa.cpl`
- Now right click on ethernet and open properties and then select Internet Protocol Version
- Then in this choose use the following DNS Server address.
- And Enter preferred DNS Server as AD-Server ip address.
- Same thing do in server 2.
- Now open AD-server Machine



- Now click on the Flag and then click Promote this server to a domain controller.

Deployment Configuration

TARGET SERVER
Ad-Server

Deployment Configuration

Domain Controller Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Select the deployment operation

- ☐ Add a domain controller to an existing domain
- ☐ Add a new domain to an existing forest
- ☒ Add a new forest

Specify the domain information for this operation

Root domain name:

[More about deployment configurations](#)

< Previous

Next >

Install

Cancel

- Choose Adda new Forest and Enter name riteshlab.local.
- Then click next.

Domain Controller Options

TARGET SERVER
Ad-Server

Deployment Configuration

Domain Controller Options

DNS Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Select functional level of the new forest and root domain

Forest functional level: Windows Server 2016 ▼

Domain functional level: Windows Server 2016 ▼

Specify domain controller capabilities

☒ Domain Name System (DNS) server☒ Global Catalog (GC)☐ Read only domain controller (RODC)

Type the Directory Services Restore Mode (DSRM) password

Password: ●●●●●●●●

Confirm password: ●●●●●●●●

[More about domain controller options](#)

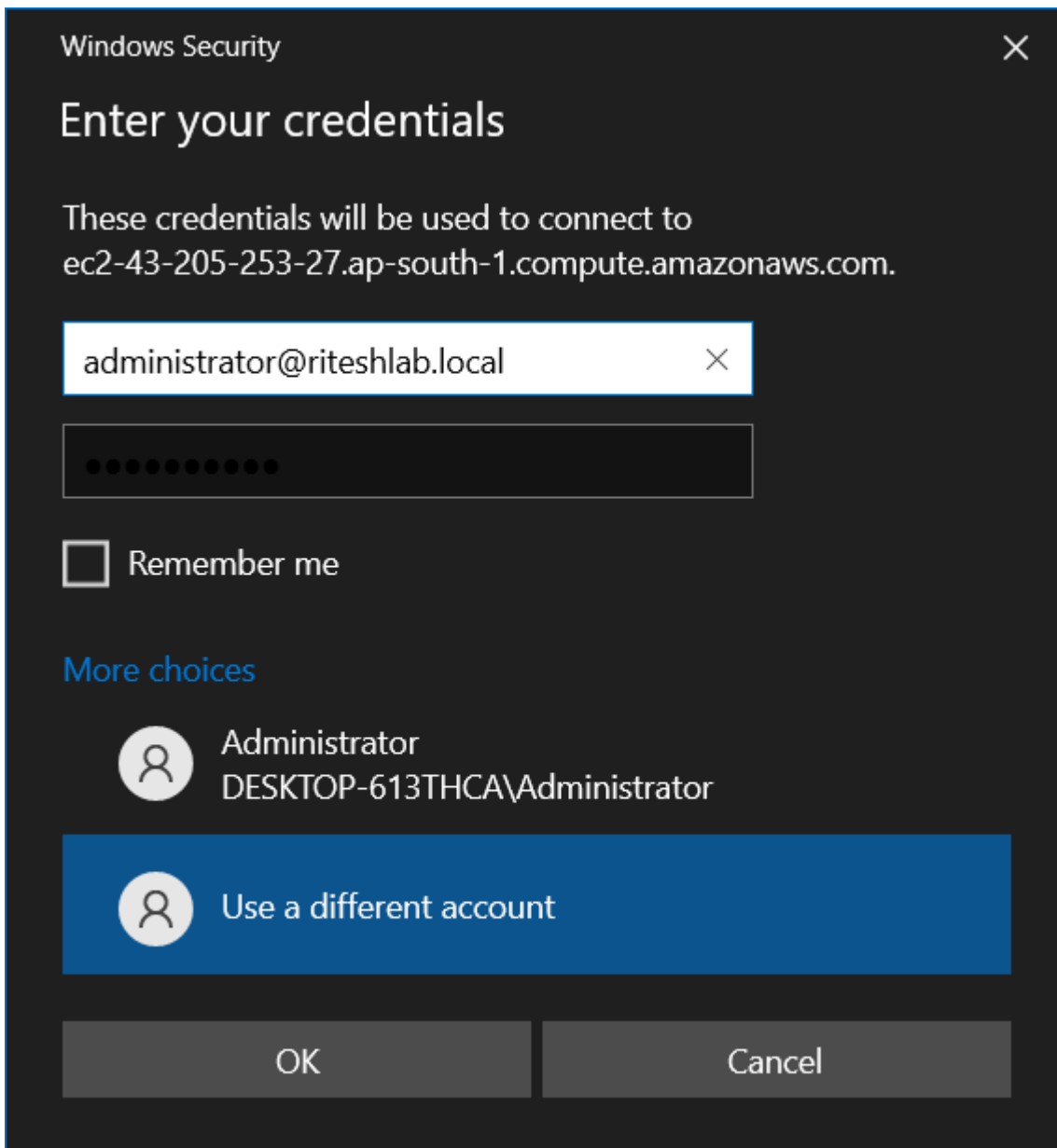
< Previous

Next >

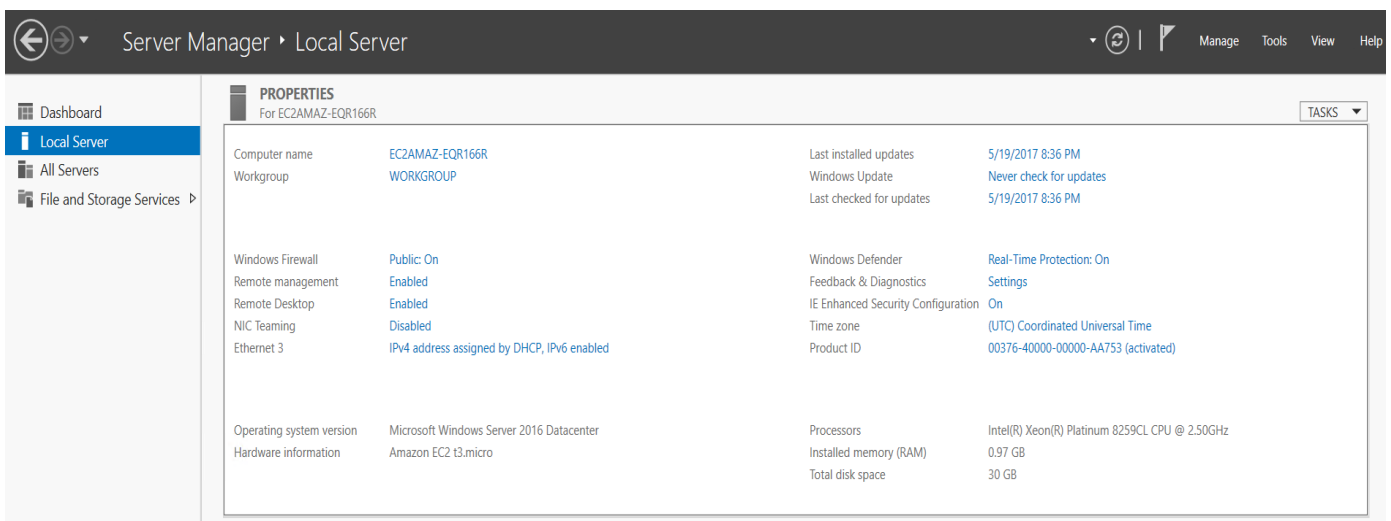
Install

Cancel

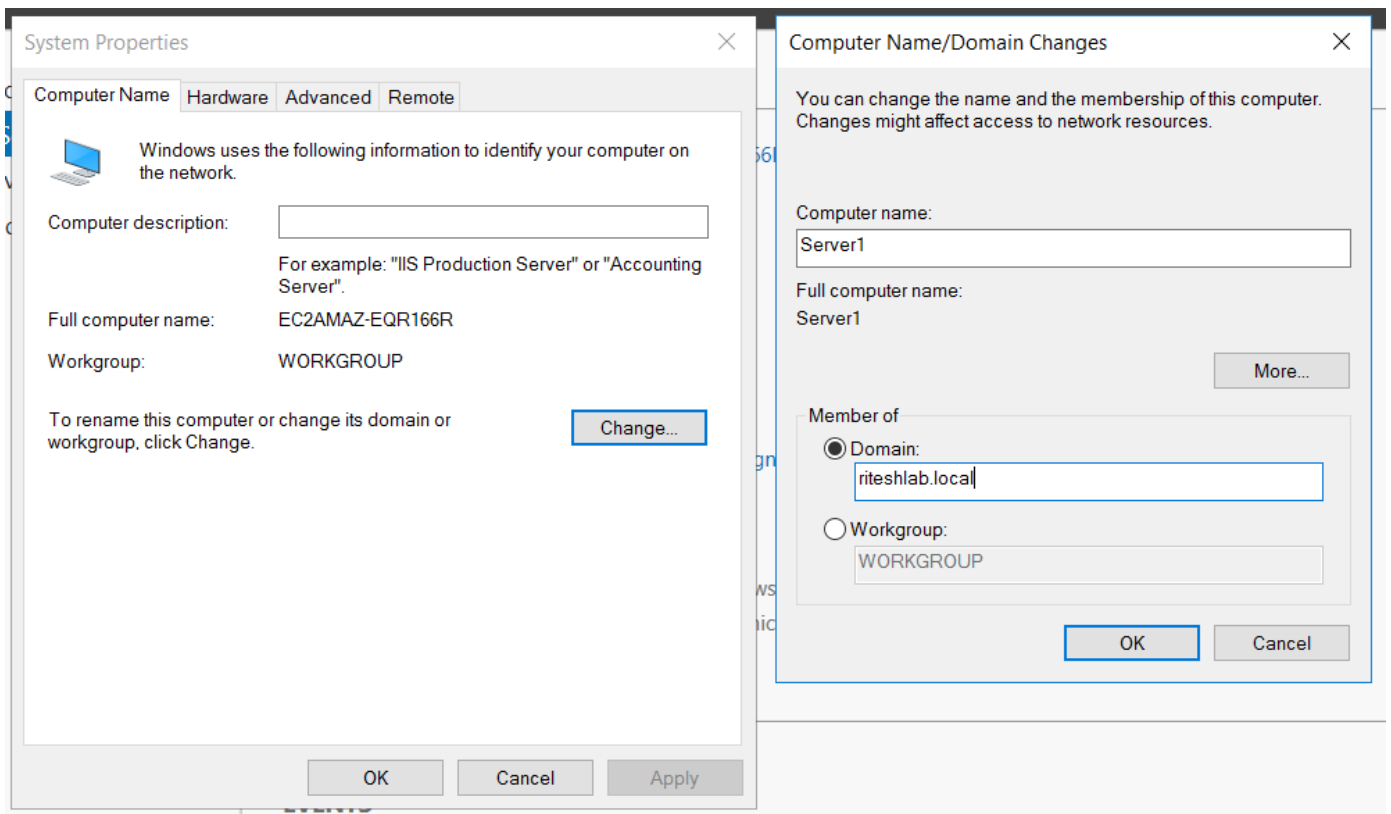
- Now Enter password.
- And then click next, next and last install.
- then it will ask for restart, restart it.
- Now again connect with the AD-Server.



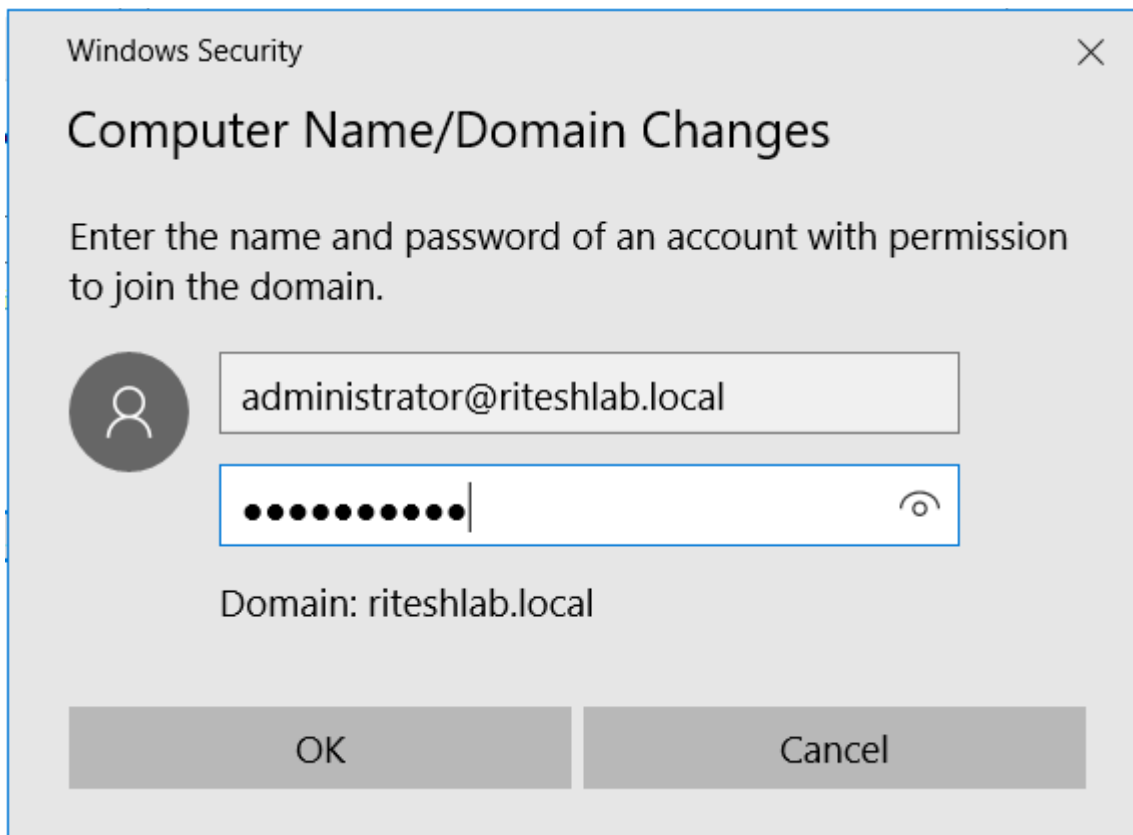
- Now we connect with that id we created normally click More choice and select use a different account.
- And Enter the created details in username add administrator@ in starting.
- After that we need to go on server 1 machine.



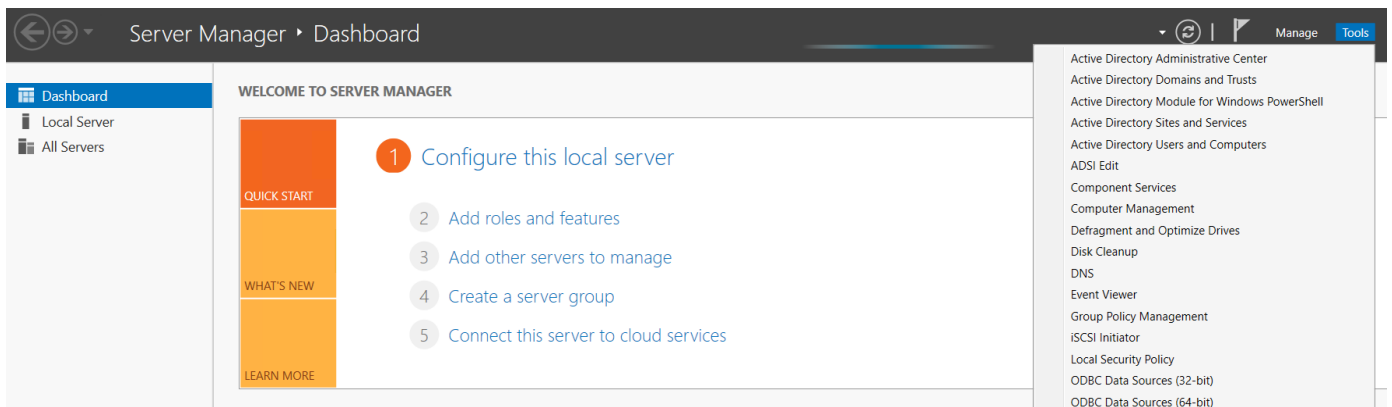
- Open server manager and select Local Server.
- Then click WORKGROUP.



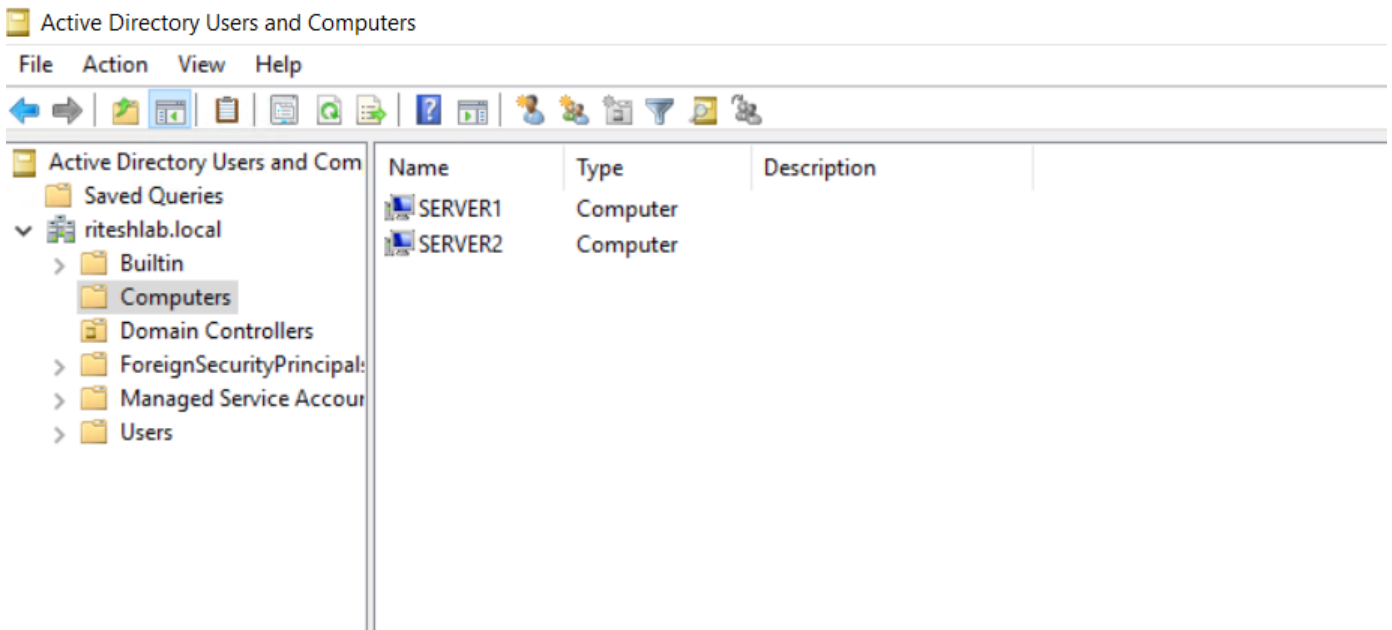
- Then click no change.
- Then give name select domain and name that you created.
- And click ok.



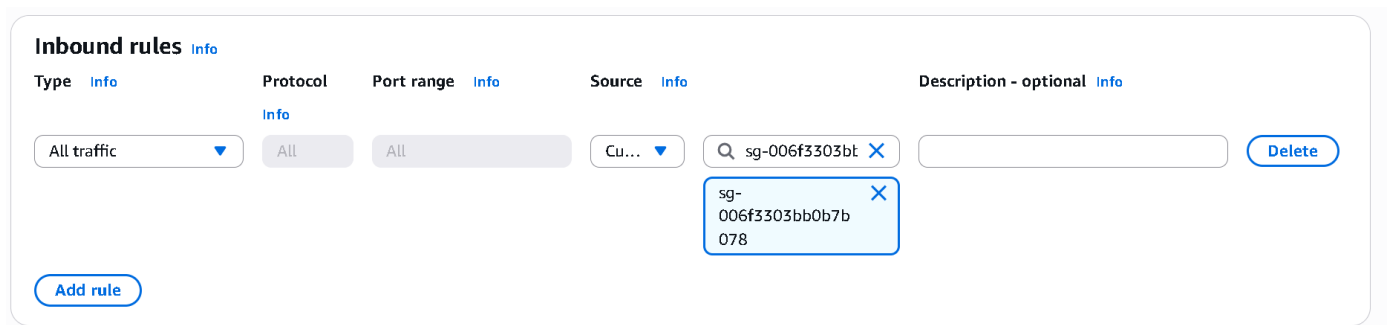
- Now this window popup Enter the domain with adding administrator@ in starting and password.
- Then click ok.
- Then restart popup come click restart now.
- Same thing do in server 2.
- Now check it add or not.
- Go to AD-server machine.



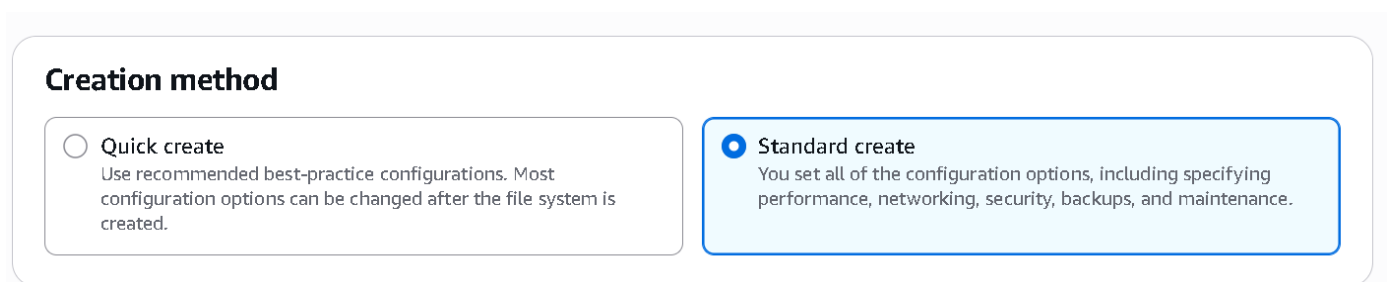
- Open Server manager and click on tool at top.
- Then select active directory users and computer.



- Now see our Server1 and server2 both are connected.
- Now we need to configure FXs so firstly I create a security group for the computer.



- Inbound rule of FxsSG Type All traffic, Source vmSG security group.
- Now we need to create FXs search on console and open it.



- Select Standard Create.

Select file system type

File system options

☐ Amazon FSx for NetApp ONTAP

FSx
Amazon FSx
for NetApp ONTAP

☐ Amazon FSx for OpenZFS

FSx
Amazon FSx
for OpenZFS

☒ Amazon FSx for Windows File Server

FSx
Amazon FSx
for Windows File Server

☐ Amazon FSx for Lustre

FSx
Amazon FSx
for Lustre

- Select Amazon FSx for Window File Server.
- Now simply scroll down and click Next.

Quick configuration

File system name - optional [Info](#)

FsxStorage

Maximum of 256 Unicode letters, whitespace, and numbers, plus + - = . _ : /

Deployment type [Info](#)

☐ Multi-AZ (Recommended)

Multi-AZ file systems are recommended for most production workloads because they have two file servers in separate Availability Zones (AZ), providing continuous availability to data and helping protect your data against instance failure and AZ disruption.

☒ Single-AZ 2

Single-AZ 2 is the latest generation of single Availability Zone file systems, and it supports SSD and HDD storage.

☐ Single-AZ 1

SSD storage capacity [Info](#)

32 GiB

Minimum 32 GiB; Maximum 65,536 GiB

- Enter name.
- Select Deployment type Single- AZ 2
- Now Select Storage capacity 32 GiB

Network & security

Virtual Private Cloud (VPC) | [Info](#)

Specify the VPC from which your file system is accessible.

vpc-072c4acd761c3b942 (CIDR: 172.31.0.0/16) ▼

VPC Security Groups | [Info](#)

Specify VPC Security Groups to associate with your file system's network interface.

Choose VPC security group(s) ▼

sg-02af4309cdb6b42d0 (FxsSG) ✕

Subnet | [Info](#)

Specify the subnet in which your file system's network interface resides.

subnet-04ffa8e5e9c79ad3d (ap-south-1a | aps1-az1) ▼

- Now select Security group.
- Scroll down.

Windows authentication

Choose an Active Directory to provide user authentication and access control for your file system | [Info](#)

- ☐ AWS Managed Microsoft Active Directory
- ☒ Self-managed Microsoft Active Directory

Active Directory prerequisites

Before you create an Amazon FSx file system joined to your AD, please ensure that your AD domain controller and DNS server are reachable from the FSx subnet(s), and the service account has the required permissions delegated to it. If your AD has multiple sites, then please ensure that your FSx subnet(s) are associated with one specific site.

- [Required Prerequisites](#) 
- [Amazon FSx Network Validation tool](#) 

- Now Select Self-managed Microsoft Active Directory.
- Scroll down.

Active Directory domain name

This is the fully qualified domain name of your self-managed directory

riteshlab.local

DNS server IP addresses

The IP DNS server IP addresses for your domain. At least one IP address is required. Enter a comma separated list.

172.31.42.0

Credentials

You can use AWS Secrets Manager or enter your credentials.

☐ **Stored in AWS Secrets Manager - *most secure***
Username and Password are stored in AWS Secrets Manager.

☒ **Provide Credentials**
Enter Username and Password.

- Enter domain name that we created in AD-Server.
- Then Enter IP address that we get in AD-Server CMD.

- Then select Service account username

Service account username

The username of the service account your resource will use to join your Active Directory. Do not include a domain prefix or suffix. For example, for 'EXAMPLE/ADMIN', exclude the domain prefix 'EXAMPLE'.

Service account password

Provide the password for the service account you are using to join this resource to your Active Directory.

Maximum of 128 characters.

Confirm password

- Now Enter Administrator Details.
- Now scroll down and click Next.

Tags

< 1 >

Tag key

Value

You don't have any tags.

Cancel

Back

Create file system

- Now Simply, scroll down and click Create file System.
- Now open all Servers.

File systems (1)



Attach

Actions ▼

Create file system

Filter file systems by property or tag

< 1 >



	File system name ▼	File system ID ▼	File system type ▼	Status ▼	Deployment type ▼	Storage class ▼	St ca
☒	FsxStorage	fs-064ea5ebc65f54496 	Windows	Available	Single-AZ 2	SSD	32

- It's created now select it and click Attach.

▼ From Windows instances (Amazon EC2, Amazon WorkSpaces, VMware Cloud on AWS)

▼ Prerequisites

1. Join an EC2 Windows instance to your Active Directory (riteshlab.local)
 - [Launch new EC2 Windows instance joined to ActiveDirectory](#) 
 - [Manually join an existing EC2 Windows instance to ActiveDirectory](#) 
2. [Connect to EC2 Windows instance](#) 

▼ Attach instruction - using the default DNS name

1. Open a command prompt.
2. Execute the following command (you can replace "Z:" with any other available drive letter):

```
net use Z: \\amznfsxjeooheu.riteshlab.local\share
```

- Now copy this command and paste in CMD of server 1 and 2

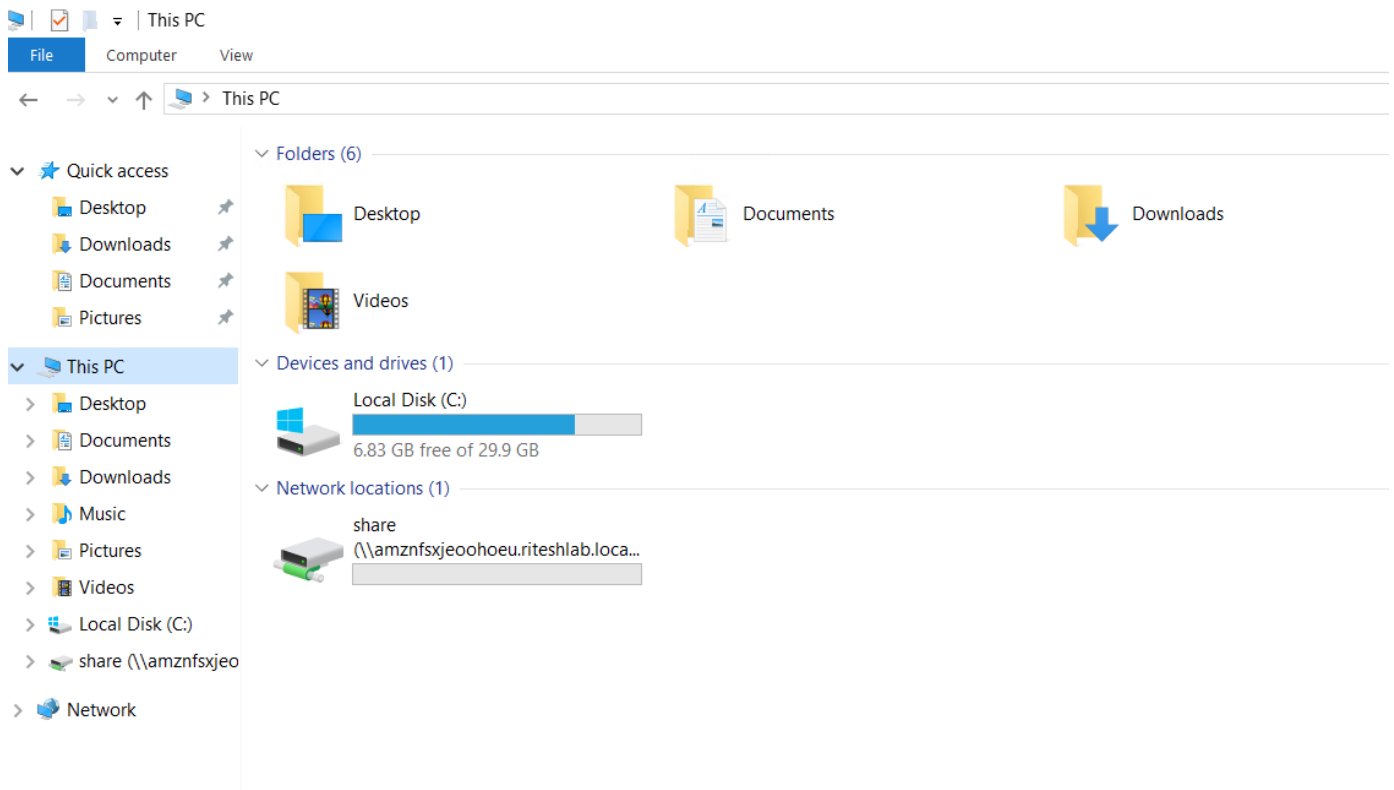
Administrator: C:\Windows\system32\cmd.exe

```
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

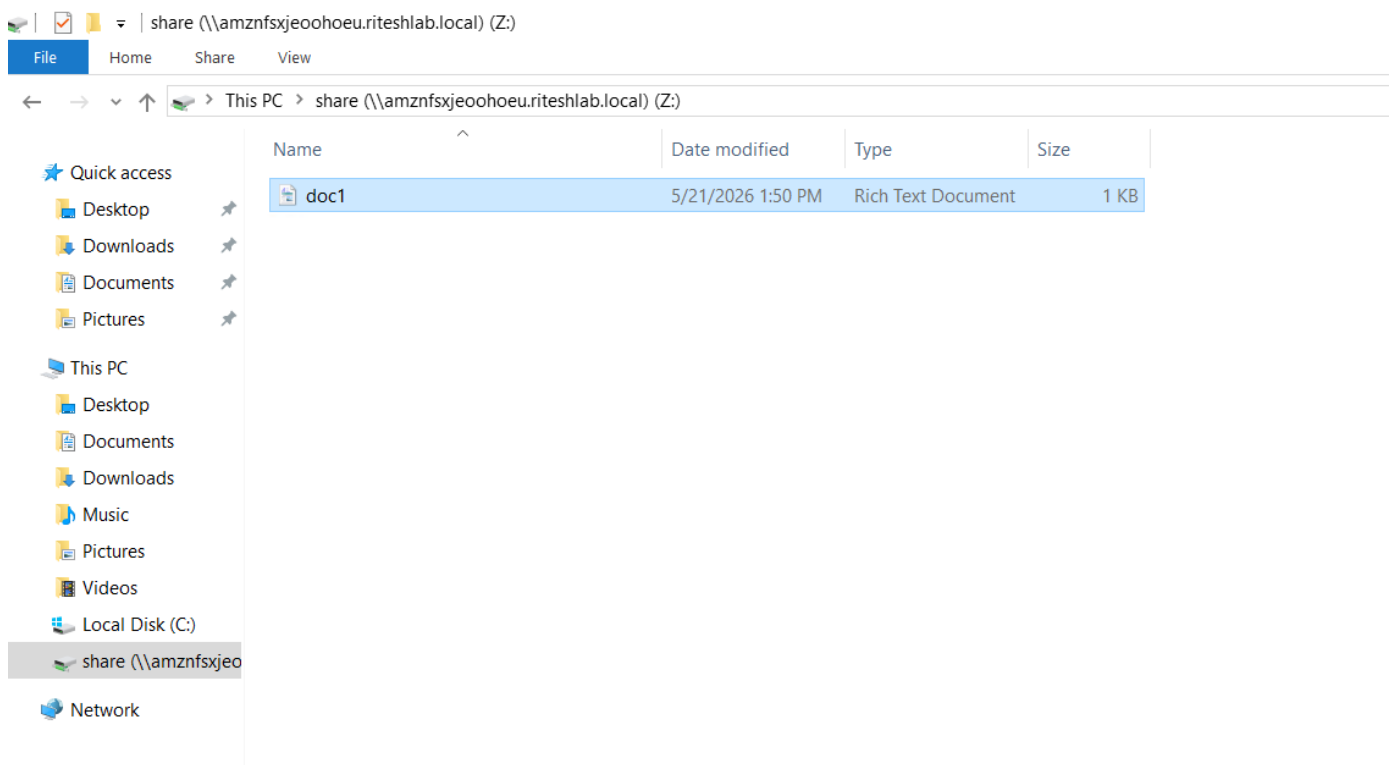
C:\Users\Administrator.RITESH LAB>net use Z: \\amznfsxjeooheu.riteshlab.local\share
The command completed successfully.

C:\Users\Administrator.RITESH LAB>
```

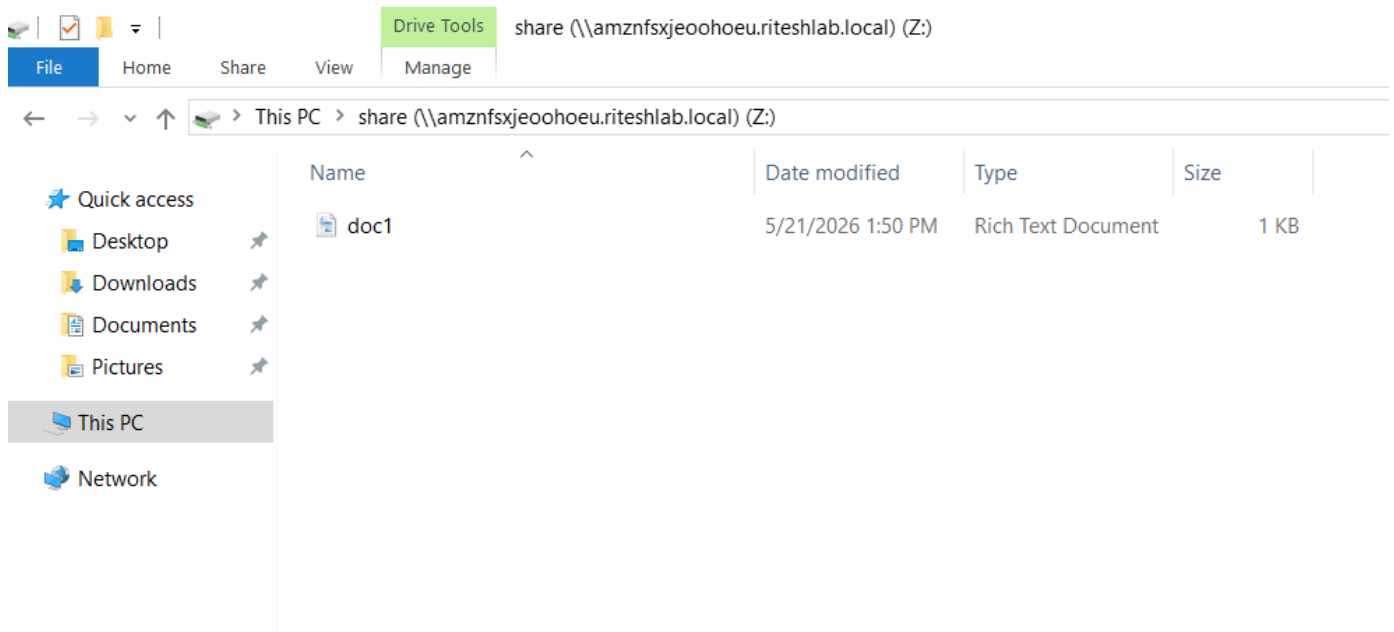
- Now we go to this pc in Server 1.



- Now open this share disk.



- Now I created one file doc1.
- Now go to server 2 in this pc.



- Now see this file is also present in Server 2.
- So this is the one kind of shared storage.
- If I am going to delete this file from here okay.
- Now I am not able to get it from the server one as well.